Chapter 5 Assignment — Trigonometric Functions: Unit Circle Approach

Name: ____________________ Date: ____________________

How to use this set: Each topic begins with *Strategy Notes* and a *Worked Example.* Then try the *Practice* (skill) and the *Challenge* (deeper) items. Give exact answers in radicals and π when possible. Clearly state domain restrictions when relevant.

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1. **Topic 1: The Unit Circle and Quadrants**

Strategy Notes. Use reference angles and quadrants to determine signs. On the unit circle, $P(\theta) = (\cos\theta, \sin\theta)$; special angles $30°, 45°, 60°$ (or $\frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}$) generate exact values.

Worked Example. Determine the sign of $\cot 225°$ and evaluate $\sin\left(-\frac{\pi}{3}\right)$.

Solution. Note $225° = 180° + 45°$, which lies in quadrant III where both $\sin\theta$ and $\cos\theta$ are negative. Since $\cot\theta = \dfrac{\cos\theta}{\sin\theta}$, the ratio of two negatives is positive, and neither numerator nor denominator is zero at a $45°$ reference angle. Hence $\cot 225°$ is *positive*. For the second part, use odd symmetry of sine: $\sin(-\theta) = -\sin\theta$. Because $\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$, we have $\sin\left(-\frac{\pi}{3}\right) = -\frac{\sqrt{3}}{2}$.

Practice.

(a) Classify as positive, negative, zero, or undefined: $\sin 301°$, $\cos 118°$, $\tan\left(-\frac{7\pi}{6}\right)$, $\csc\left(\frac{\pi}{2}\right)$.

(b) Find exact coordinates of the terminal point on the unit circle for $\theta = \frac{5\pi}{4}$ and for $\theta = -\frac{2\pi}{3}$.

Challenge.

(a) An angle θ has $\cos\theta = -\frac{2}{3}$ and lies in quadrant II. Find $\sin\theta$ and $\tan\theta$ exactly.

2. **Topic 2: Trigonometric Functions of Real Numbers**

   Strategy Notes. Use periodicity and reference angles: $\sin(\theta + 2\pi k) = \sin\theta$, $\cos(\theta + 2\pi k) = \cos\theta$, $\tan(\theta + \pi k) = \tan\theta$. For counts on intervals, convert to a base solution then track period.

   Worked Example. How many $\theta \in [0, 4\pi]$ satisfy $\sin\theta = \frac{\sqrt{2}}{2}$?

   Solution. In $[0, 2\pi)$ the equation $\sin\theta = \frac{\sqrt{2}}{2}$ holds at the reference-angle positions $\theta = \frac{\pi}{4}$ and $\theta = \frac{3\pi}{4}$. Over the interval $[0, 4\pi]$ we include one more full cycle, so add 2π to each value to obtain $\theta = \frac{9\pi}{4}$ and $\theta = \frac{11\pi}{4}$. Therefore there are **four** solutions: $\{\frac{\pi}{4}, \frac{3\pi}{4}, \frac{9\pi}{4}, \frac{11\pi}{4}\}$.

   Practice.

   (a) Evaluate exactly: $\csc\left(-\frac{7\pi}{6}\right)$ and $\cos\left(\frac{11\pi}{3}\right)$.

   (b) How many solutions are in $0 \le \theta \le 3\pi$ for $\cos\theta = -\frac{1}{2}$? List them.

3. **Topic 3: Graphs of** sin **and** cos

Strategy Notes. For $y = A\sin(Bx - C) + D$ (or cosine) amplitude $= |A|$, period $= \frac{2\pi}{|B|}$, phase shift $= \frac{C}{B}$ to the right, and midline $y = D$.

Worked Example. Describe and sketch $y = -3\sin\left(2x - \frac{\pi}{4}\right) + 2$.

Solution. Write the function in the form $y = A\sin(Bx - C) + D$ with $A = -3$, $B = 2$, $C = \frac{\pi}{4}$, and $D = 2$.

- Amplitude $= |A| = 3$.
- Period $= \dfrac{2\pi}{|B|} = \dfrac{2\pi}{2} = \pi$.
- Phase shift $= \dfrac{C}{B} = \dfrac{\pi/4}{2} = \frac{\pi}{8}$ to the right.
- Midline $y = D = 2$ and range $[2 - 3,\ 2 + 3] = [-1, 5]$.

One convenient window for a single period is from $x = \frac{\pi}{8}$ to $x = \frac{\pi}{8} + \pi$. Because $A < 0$, the graph starts at a *minimum* at $x = \frac{\pi}{8}$, rises to the midline at $x = \frac{\pi}{8} + \frac{\pi}{4}$, reaches a maximum at $x = \frac{\pi}{8} + \frac{\pi}{2}$, returns to the midline at $x = \frac{\pi}{8} + \frac{3\pi}{4}$, and completes the period at a minimum at $x = \frac{\pi}{8} + \pi$.

Practice.

(a) For $f(x) = 2\sin\left(4x + \frac{\pi}{3}\right) - 1$, state amplitude, period, phase shift, midline, and range. Sketch one period.

(b) Write an explicit sine or cosine function with amplitude 3, period π, midline $y = -2$, and a phase shift of $\frac{\pi}{8}$ to the right.

Challenge.

(a) The function reaches a maximum of 5 at $x = \frac{\pi}{12}$ and a minimum of -1 at $x = \frac{5\pi}{12}$. Assuming it is sinusoidal with least positive period, find a formula.

4. **Topic 4: Graphs of** tan**,** cot**,** sec**, and** csc

Strategy Notes. For $y = a\tan(bx - c)$, period $= \frac{\pi}{|b|}$ and vertical asymptotes where $bx - c = \frac{\pi}{2} + k\pi$. For $y = a\sec(bx - c) + d$, identify the underlying cosine to locate asymptotes and vertices.

Worked Example. Determine the period and two consecutive vertical asymptotes for $y = -3\tan\big(2x - \frac{\pi}{2}\big)$.

Solution. For $y = a\tan(bx - c)$ the period is $\dfrac{\pi}{|b|}$. Here $b = 2$, so the period is $\frac{\pi}{2}$. Vertical asymptotes occur where the tangent input equals $\frac{\pi}{2} + k\pi$:

$$2x - \tfrac{\pi}{2} = \tfrac{\pi}{2} + k\pi \quad\Rightarrow\quad 2x = \pi + k\pi \quad\Rightarrow\quad x = \tfrac{\pi}{2}(k + 1).$$

Taking $k = 0$ and $k = 1$ gives two consecutive asymptotes at $x = \frac{\pi}{2}$ and $x = \pi$.

Practice.

(a) For $g(x) = -3\tan\big(2x + \frac{\pi}{2}\big)$, give the period, range, and two consecutive vertical asymptotes.

(b) What is the period of $h(x) = (\sin x)(\cos x)$? Justify.

Challenge.

(a) Graph one period of $y = 1 + 2\sec(\frac{x}{2})$. Indicate asymptotes and key points.

5. **Topic 5: Inverse Trigonometric Functions**

   Strategy Notes. Principal-value ranges: $\arcsin x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, $\arccos x \in [0, \pi]$, $\arctan x \in (-\frac{\pi}{2}, \frac{\pi}{2})$. Use right-triangle or unit-circle triangles to rewrite expressions like $\sin(\arctan x)$ in algebraic form.

   Worked Example. Let $y = \arctan x$. Find $\cos y$ in terms of x.

   Solution. If $y = \arctan x$, then $\tan y = x$. Interpret this as a right triangle with opposite side x and adjacent side 1. By the Pythagorean theorem, the hypotenuse is $\sqrt{1+x^2}$. Therefore $\cos y = \dfrac{\text{adjacent}}{\text{hypotenuse}} = \dfrac{1}{\sqrt{1+x^2}}$. Since $y \in (-\frac{\pi}{2}, \frac{\pi}{2})$, this value is nonnegative as expected.

   Practice.

   (a) Evaluate: $\arccos(-\frac{1}{2})$, $\arcsin(1)$, $\arctan(0)$.

   (b) Let $y = \arctan x$. Express $\sin y$ and $\sec y$ in terms of x.

Challenge.

(a) Solve $\arcsin(2t-1) = \frac{\pi}{6}$ for t, and then find all $x \in [0, 2\pi)$ satisfying $\sin x = 2t - 1$.

6. **Topic 6: Modeling Harmonic Motion**

Strategy Notes. A sinusoidal model has the form $y = A\sin(\omega t - \phi) + D$ or $A\cos(\omega t - \phi) + D$. Amplitude $= |A|$, midline $y = D$, and period $T = \frac{2\pi}{\omega}$. From "high-to-low" time you can infer half the period.

Worked Example. A weight oscillates between 1.0 m and 1.6 m from the ceiling, reaching a high at $t = 0$ and a low after 0.9 s. Build a model.

Solution. The maximum and minimum are 1.6 and 1.0, so $A = \dfrac{\max - \min}{2} = \dfrac{1.6 - 1.0}{2} = 0.3$ and the midline is $D = \dfrac{\max + \min}{2} = \dfrac{1.6 + 1.0}{2} = 1.3$. From high to low is half a period, so $\frac{T}{2} = 0.9 \Rightarrow T = 1.8$ and $\omega = \dfrac{2\pi}{T} = \dfrac{2\pi}{1.8}$. Because the mass is at a *high* at $t = 0$, a cosine model with zero phase works:

$$d(t) = 1.3 + 0.3\cos\left(\tfrac{2\pi}{1.8}t\right).$$

Practice.

(a) A spring-mass moves between 1.1 m and 1.9 m from the ceiling. It takes 1.4 s to go from low to high. At $t = 0$ it is at the midline moving upward. Build a model $d(t)$ and find $d(2.4)$.

(b) A Ferris wheel has radius 12 m and rotates once every 28 s. The lowest seat is 3 m above the ground and at $t = 0$ the seat is at the lowest point. Write the height function $h(t)$.

Challenge.

(a) A sinusoid has midline $y = 4$, amplitude 1.5, and period 6. The graph passes through $(1, 4)$ with positive slope. Find a formula $y(t)$.